Week 8 BA

2024 Sep01 ET

1. [Source Q5; MCQ]

Assume 5 factories are producing the same output of 1000 units. But, the resources used i.e. labour and raw materials are different. Which of the following factories are efficient?

Factory	Labour	Raw Material
Factory A	100	171
Factory B	105	175
Factory C	108	162
Factory D	104	165
Factory E	109	170

(a) A, B, D

(b) B, C, D

(c) A, C, D

(d) A, E, D

(e) B, D, E

[Source ans: c]

2. [Source Q9; MSQ]
Which of the following are true:

(a) Productive efficiency frontiers are all combinations of outputs such that theproduction of one unit cannot be increased without sacrificing without sacrificing the other.

(b) Organizations that do not find themselves on the Economic frontier are calledinefficient economic units or organizations.

(c) Productive efficiency focuses on maximizing the given output under givenconstraints without focusing on optimally allocating the products.

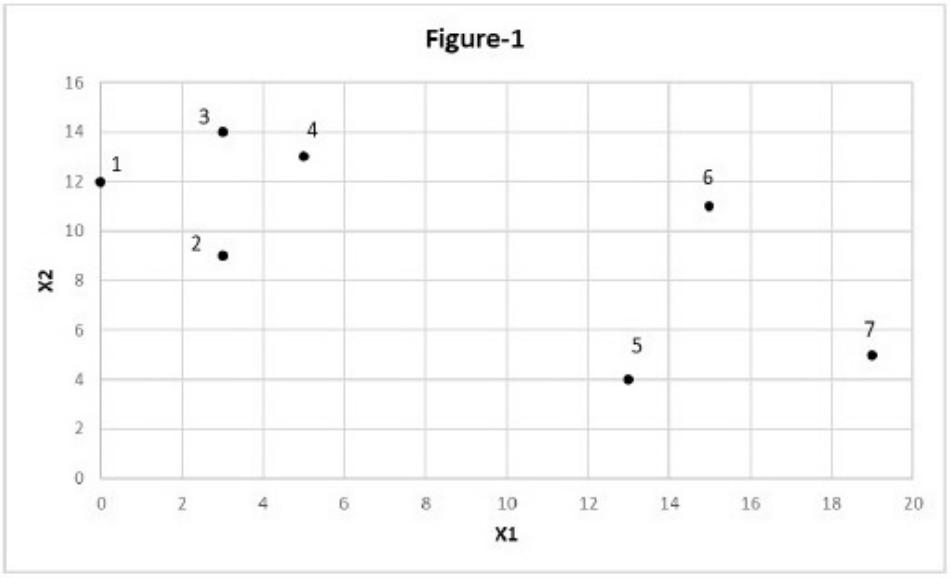
(d) DEA focuses on the technology to improve the productive efficiency.

[Source ans: a, b, c]

2024 Dec01 Quiz-2

Comprehension for questions 3-4 (source Q19)

You are provided the below (Figure-1) mapping for a DEA problem. With this answer the given subquestions



3. [Source Q20; MCQ]
If the axis "X1" and "X2" represent the inputs for the DEA problem, then which of the following would you assume/guess is the efficient frontier (if the output is going to be held constant for all DMUs)?

(a) 1-2-5-7

(b) 1-3-4-6-7

(c) 3-2-5-6-4

(d) All of these

[Source ans: a]

4. [Source Q21; MCQ]
If the axis "X1" and "X2": represent the outputs for the DEA problem, then which of the following would you assume/guess is the efficient frontier (if the input is going to be held constant for all DMUs)?

(a) 1-2-5-7

(b) 1-3-4-6-7

(c) 3-2-5-6-4

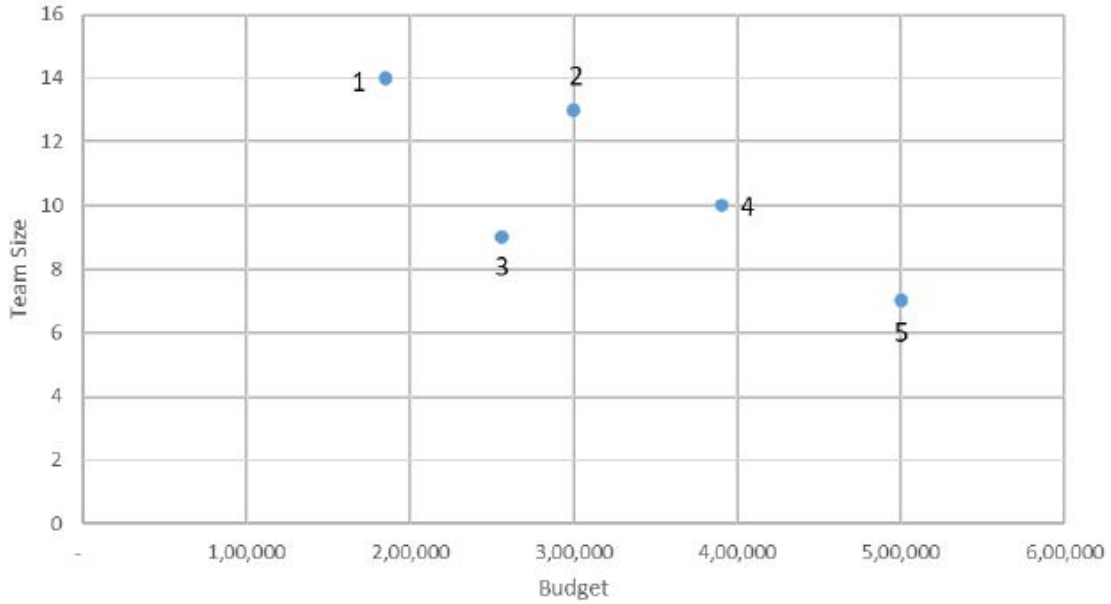
(d) All of these

[Source ans: b]

2024 Dec22 ET

Comprehension for questions 5-7 (source Q27)

As a Business Analyst, we are trying to find the DMUs that are efficient, where there are 2 inputs (Team Size & Budget) and one constant output. Based on the graph given below, answer the subquestions



5. [Source Q28; MCQ]
Which DMUs are in the economic frontier?

(a) (1,2,4)

(b) (1,3,5)

(c) (1,2,4,5)

(d) (1,3,4)

[Source ans: b]

6. [Source Q29; MCQ]
Which DMUs are the reference units for DMU 2?

(a) (1,4)

(b) (1,3)

(c) (3,5)

(d) (3,4)

[Source ans: b]

7. [Source Q30; MCQ]
Which DMUs are the reference units for DMU 4?

(a) (1,4)

(b) (1,3)

(c) (3,5)

(d) (3,4)

[Source ans: c]

2025 Apr13 ET

Comprehension for questions 8-10 (source Q30)

As a Business Analyst, we are trying to find the DMUs that are efficient, where there are 2 outputs (Number of Leads & Sales) and one constant input. Based on the graph given in Figure-4 below, answer the given sub-questions

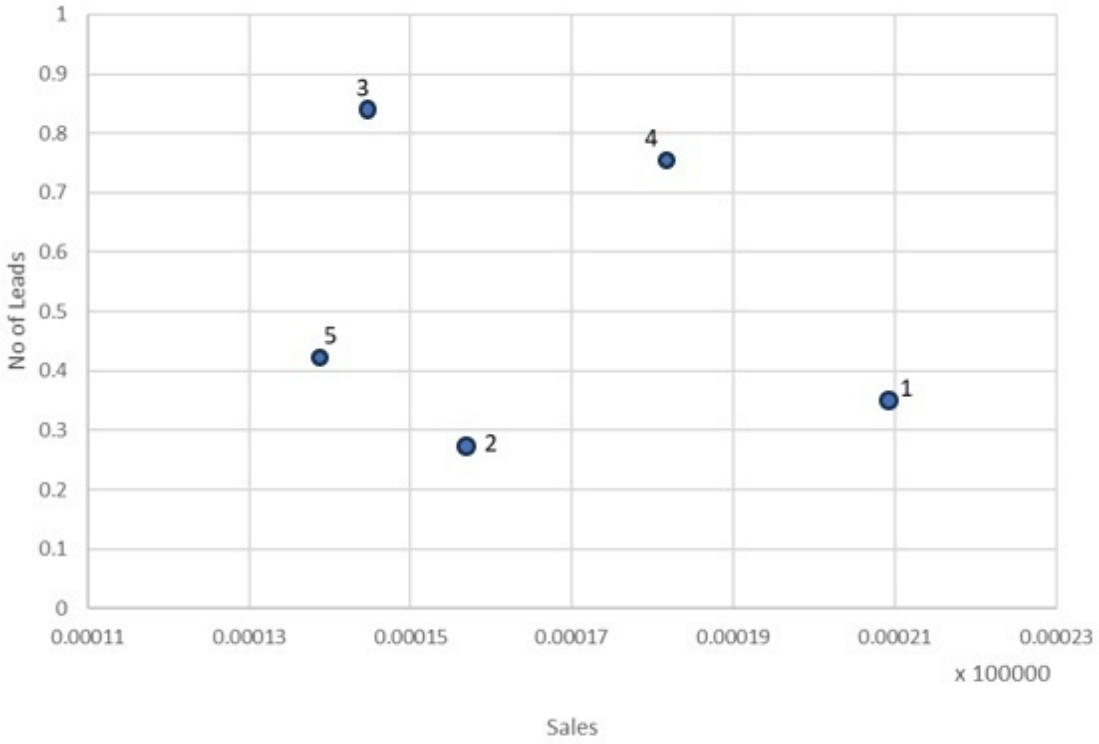


Figure-4

8. [Source Q31; MCQ]
Which DMUs are in the economic frontier?

(a) (3,2,1)

(b) (3,2,5)

(c) (4,5,2)

(d) (3,4,1)

[Source ans: d]

9. [Source Q32; MCQ]
Which DMUs are the reference units for DMU 2?

(a) (5,4)

(b) (4,1)

(c) (3,5)

(d) (3,1)

[Source ans: b]

10. [Source Q33; MCQ]
Which DMUs are the reference units for DMU 5?

(a) (1,4)

(b) (1,3)

(c) (3,4)

(d) None of these

[Source ans: c]

2026 May06 ET

Comprehension for questions 11-13 (source Q1)

The following scenario is purely hypothetical A data science company has 5 managers (5 DMUs and k=5). Each manager is allocated the same budget of Rs.3,00,000. The savings achieved (in INR) and the projects completed (successfully) are the outputs.

Manager	Savings	Projects Completed
1	13,50,000	14
2	18,50,000	7
3	32,50,000	14
4	13,24,000	27
5	25,00,000	23

Table 1: Manager Performance

Formulate an optimization problem for Manager 2 to calculate his/her efficiency. Here, the only input is (I_{ik}): the budget to run all the projects. Corresponding to this, there is one input weight (x_{ik}). The outputs are (O_{jk}): Savings, and the Projects Completed. The corresponding output weights are denoted by (y_{jk}).

The LP formulation of the DEA problem for Manager 2.

$$\begin{gathered} Max\ \ a * 18{,}50{,}000 + b * 7 \\ subject\ to\ \ a * 3{,}00{,}000 = 1 \\ a * 13{,}50{,}000 + b * 14 \leq c * 3{,}00{,}000 \\ a * 18{,}50{,}000 + b * 7 \leq c * 3{,}00{,}000 \\ a * 32{,}50{,}000 + b * 14 \leq c * 3{,}00{,}000 \\ a * 13{,}24{,}000 + b * 27 \leq c * 3{,}00{,}000 \\ a * 25{,}00{,}000 + b * 23 \leq c * 3{,}00{,}000 \\ Decision\ variables:\ a, b, c \geq 0 \end{gathered}$$

Based on the data provided, answer the given sub-questions.

11. [Source Q5; MCQ]

Which DMUs are in the economic frontier?

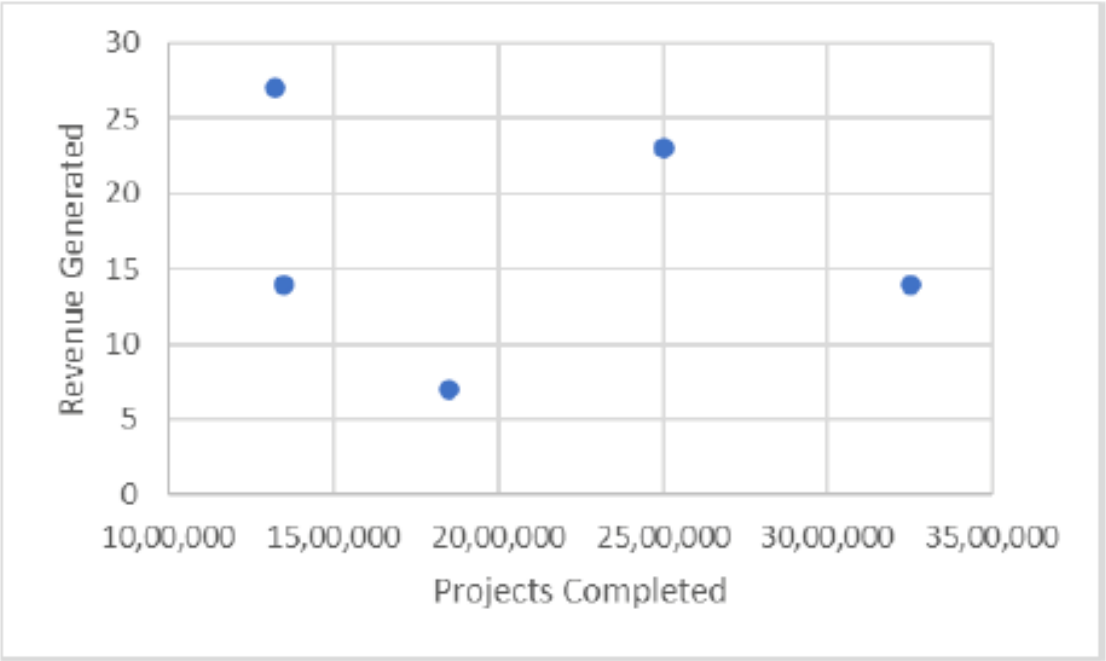


Figure 1: Manager Performance

(a) (1,2,4)

(b) (3,2,5)

(c) (1,4,5)

(d) (4,5,3)

[Source ans: d]

12. [Source Q6; MCQ]

Which DMUs are the reference units for DMU 2?

(a) (5,4)

(b) (5,3)

(c) (3,5)

(d) (3,1)

[Source ans: b]

13. [Source Q7; MCQ]

Which DMUs are the reference units for DMU 1?

(a) (1,4)

(b) (1,3)

(c) (4,5)

[Source ans: c]